Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

Data Visualization with Power BI and Tableau

Practical – VIII

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Date of Assignment: 27-08-2026	Date/Time of Submission:27-08-2026

AIM :- Performing Advanced Analytics using Trends, Clustering, and Forecasting

DATASET :- Indian Climate Dataset (2024–2025)

a) Add and interpret trend lines (linear, exponential)

b) Customize and analyze trend models

c) Perform clustering on datasets

d) Analyze distributions using built-in tools

e) Apply forecasting techniques

f) Compare different statistical models

g) Export and interpret statistical results

A) Add and interpret trend lines (linear, exponential)

Task 1: Create Temperature vs Humidity Trends

Step 1: Open Tableau Desktop.

Step 2: Connect to: `Indian_Climate_Dataset_2024_2025.csv`

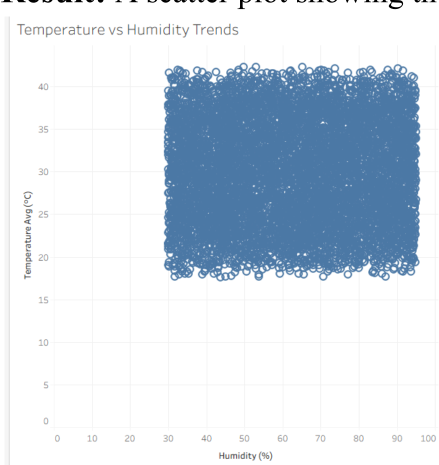
Step 3: Create a new worksheet. Rename it: Temperature vs Humidity Trends

Step 4: Drag: `Humidity (%)` → `Columns`

Step 5: Drag: `Temperature_Avg (°C)` → `Rows`

Step 6: Go to: Analysis → uncheck Aggregate Measures. This plots individual observations instead of aggregated values.

Result: A scatter plot showing the relationship between humidity and average temperature.



Task 2: Incorporating a Linear Trend Line

Step 1: Access the Analytics window.

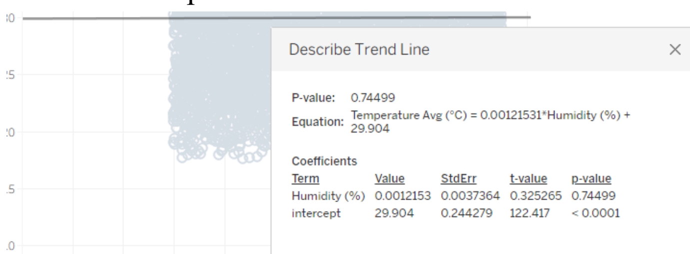
Step 2: From the Model section, select and pull: **`Trend Line`** directly onto the workspace.

Step 3: Release the cursor over: **`Linear`**

Step 4: Position the pointer over the resulting line. The application will present:

- Statistical linear formula
- R-squared coefficient
- Probability value (P-value)

Result: The visualization now includes a line demonstrating the correlation between humidity levels and temperature shifts.



Task 3: Integrating an Exponential Trend Model

Step 1: Initialize a fresh worksheet. Assign label: **Humidity vs Rainfall**

Step 2: Pull: **`Humidity (%)`** → **`Columns`**

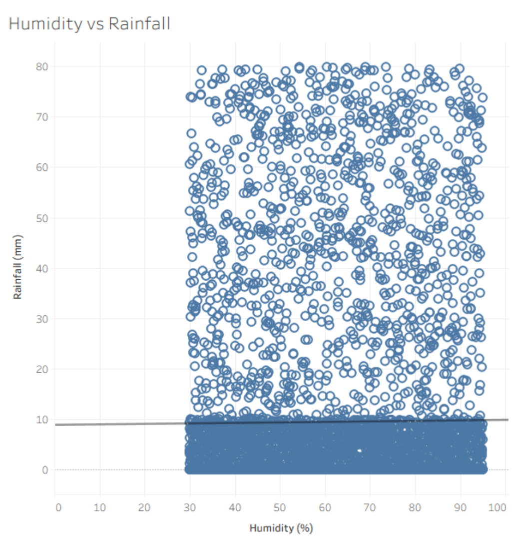
Step 3: Pull: **`Rainfall (mm)`** → **`Rows`**

Step 4: Navigate to: Analysis → deselect Aggregate Measures

Step 5: Enter the Analytics window.

Step 6: Move Trend Line onto the plot.Choose: **`Exponential`**

Result: A curve representing the non-linear correlation between moisture levels and precipitation volume.



B) Customize and analyze trend models

Task 1: Disaggregate Trend Lines by City

Step 1: Return to the worksheet labeled: **Temperature vs Humidity Trends**

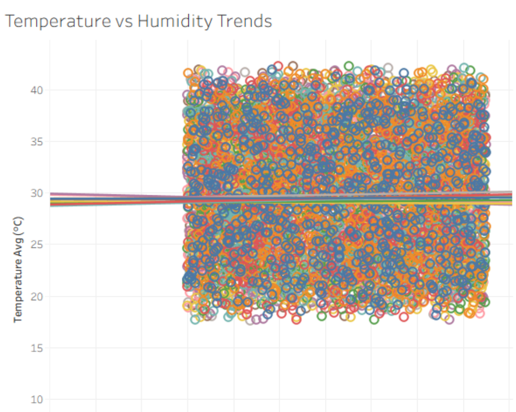
Step 2: Drag the following dimension: **`City` → `Color`**

Step 3: Place it onto the Marks card.

The software will automatically generate distinct trend lines for every city.

Result:

Each urban area now displays a unique correlation between atmospheric moisture and temperature.



Task 2: Compare Statistical Models

Step 1: Right-click any trend line. Select: **Edit Trend Lines**

Step 2: Change **Model Type** between:

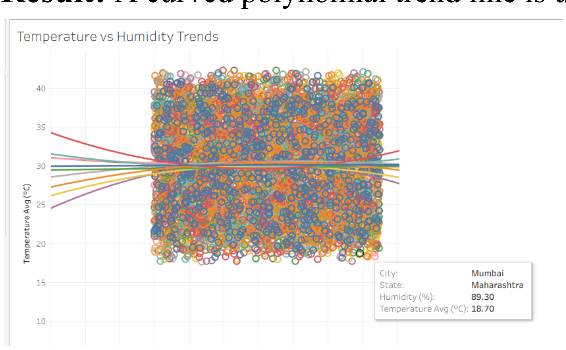
- Linear
- Logarithmic
- Exponential
- Polynomial

Step 3: Select: **Polynomial**

Step 4: Set: **Degree = 2**

Step 5: Click **OK**.

Result: A curved polynomial trend line is used to model possible seasonal relationships.



Task 3: Examine the Detailed Trend Model

Step 1: Right-click anywhere within the visualization area. Navigate to: **`Describe Trend Line`** This action displays a concise overview including the mathematical formula, R-squared value, and P-value.

Describe Trend Line

P-value: 0.593082

Equation: Temperature Avg (°C) = 0.000568353*Humidity (%)^2 + -0.0639077*Humidity (%) + 31.5382

Coefficients

Term	Value	StdErr	t-value	p-value
Humidity (%)^2	0.0005684	0.0007124	0.797746	0.425278
Humidity (%)	-0.0639077	0.0904095	-0.706869	0.479874
intercept	31.5382	2.69341	11.7094	< 0.0001

Step 2: Perform another right-click on the plot. Choose: **`Describe Trend Model`**
Analyze the following statistics:

- Analysis of Variance (ANOVA) table
- Calculated Degrees of Freedom
- Sum of Squared Errors (SSE)
- F-statistic significance

Describe Trend Model

Trend Lines Model

A polynomial trend model of degree 2 is computed for Temperature Avg (°C) given Humidity (%).

Model formula:	State*(Humidity (%)^2 + Humidity (%) + intercept)
Number of modeled observations:	7310
Number of filtered observations:	0
Model degrees of freedom:	30
Residual degrees of freedom (DF):	7280
SSE (sum squared error):	259327
MSE (mean squared error):	35.6218
R-Squared:	0.0033993
Standard error:	5.9684
p-value (significance):	0.686762

Analysis of Variance:

Field	DF	SSE	MSE	F	p-value
State	27	860.85792	31.8836	0.89506	0.620982

Individual trend lines:

Panes		Color	Line		Coefficients				
Row	Column	State	p-value	DF	Term	Value	StdErr	t-value	p-value
Temperature Avg (°C)	Humidity (%)	West Bengal	0.724418	728	Humidity (%)^2	-0.0004072	0.0007014	-0.580525	0.561741
					Humidity (%)	0.0576827	0.0888786	0.649006	0.516539
					intercept	28.5309	2.64375	10.7918	< 0.0001
Temperature Avg (°C)	Humidity (%)	Uttar Pradesh	0.86154	728	Humidity (%)^2	0.0002416	0.0007081	0.341113	0.733117
					Humidity (%)	-0.034828	0.0882439	-0.394679	0.693195
					intercept	31.0418	2.55841	12.1332	< 0.0001
Temperature Avg (°C)	Humidity (%)	Telangana	0.790766	728	Humidity (%)^2	0.0001521	0.0007003	0.217152	0.828151
					Humidity (%)	-0.0268618	0.0880446	-0.305093	0.760382
					intercept	31.0118	2.60546	11.9026	< 0.0001
Temperature Avg (°C)	Humidity (%)	Tamil Nadu	0.0376765	728	Humidity (%)^2	-0.0016531	0.000676	-2.44545	0.014703
					Humidity (%)	0.198202	0.0854157	2.32044	0.0205924
					intercept	24.4934	2.51522	9.73808	< 0.0001
Temperature Avg (°C)	Humidity (%)	Rajasthan	0.366146	728	Humidity (%)^2	-0.0010141	0.0007159	-1.41646	0.157069
					Humidity (%)	0.125697	0.0901763	1.3939	0.163773
					intercept	26.1069	2.65435	9.83551	< 0.0001
Temperature Avg (°C)	Humidity (%)	Maharashtra	0.927117	728	Humidity (%)^2	-7.464e-06	0.0007054	-0.0105812	0.99156
					Humidity (%)	0.0054906	0.0897737	0.0611608	0.951248
					intercept	29.4592	2.66492	11.0544	< 0.0001
Temperature Avg (°C)	Humidity (%)	Madhya Pradesh	0.593082	728	Humidity (%)^2	0.0005684	0.0007124	0.797746	0.425278
					Humidity (%)	-0.0639077	0.0904095	-0.706869	0.479874
					intercept	31.5382	2.69341	11.7094	< 0.0001
Temperature Avg (°C)	Humidity (%)	Karnataka	0.127979	728	Humidity (%)^2	0.0013417	0.0007287	1.84131	0.0659828
					Humidity (%)	-0.158472	0.0924042	-1.71499	0.0867729

C) Perform clustering on datasets

Task 1: Generate Climate & AQI Groupings

Step 1: Initialize a fresh worksheet. Assign label: **Climate & AQI Clusters**

Step 2: Pull: **`AQI` → `Columns`**

Step 3: Pull: **`Temperature_Avg (°C)` → `Rows`**

Step 4: Navigate to: **Analysis → deselect Aggregate Measures**

Step 5: Move the following dimension: **`City`** → **`Detail`** onto the Marks card.

Task 2: Implementing Clustering Techniques

Step 1: Enter the Analytics window.

Step 2: From the Model section, pull: **`Cluster`** directly onto the workspace.

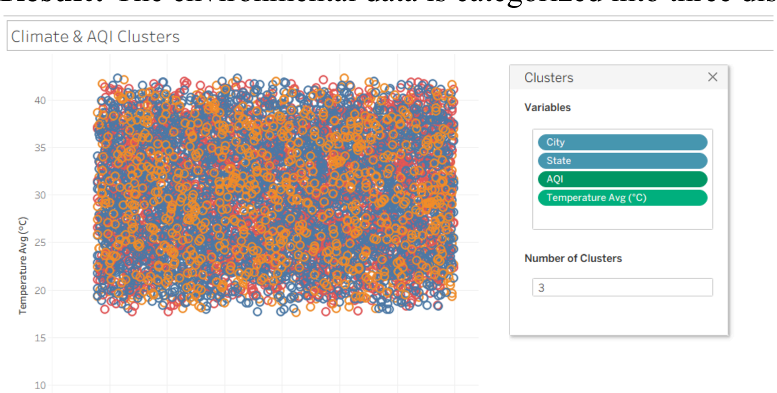
Step 3: Within the Clusters window, incorporate the following variables:

- AQI
- Temperature_Avg (°C)
- Humidity (%)
- Wind_Speed (km/h)

Step 4: Adjust the configuration to: **Number of Clusters = 3**

Step 5: Exit the dialog box. The application will automatically assign colors to data points based on their respective groups.

Result: The environmental data is categorized into three distinct statistical climate profiles.

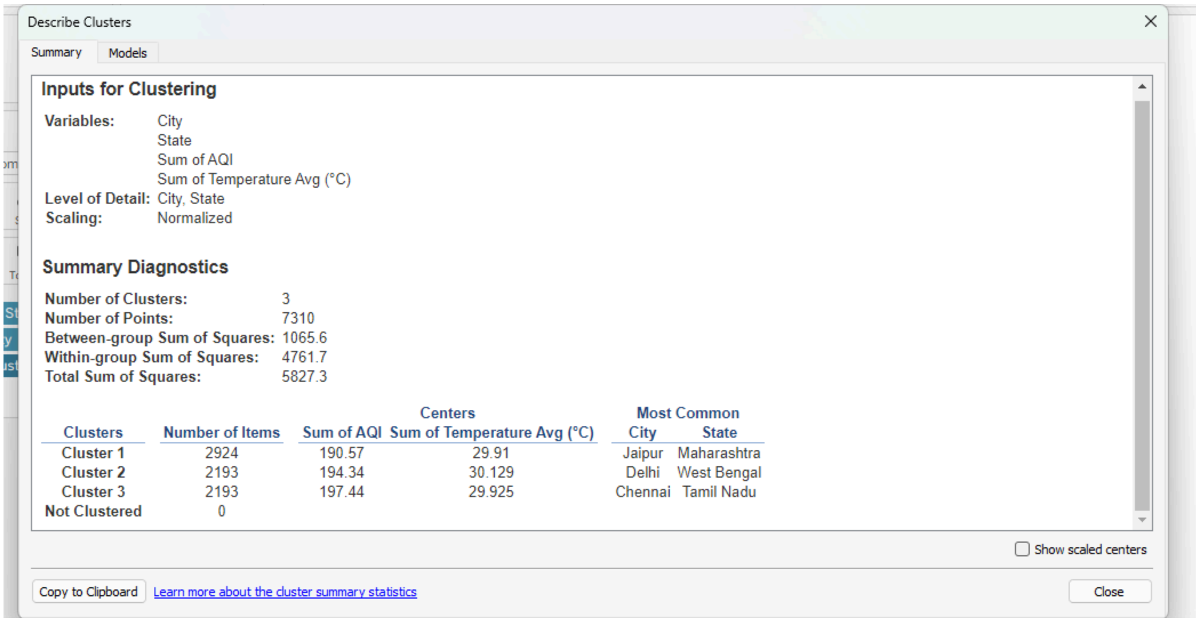


Task 3: Analyze the Grouping Characteristics

Step 1: Perform a right-click on the **`Clusters`** dimension.

Step 2: Choose the following option: **`Describe Clusters`**

Step 3: Examine the **Summary** tab to evaluate average values for AQI, temperature, moisture, and wind velocity.



Describe Clusters

Summary | Models

Inputs for Clustering

Variables: City
State
Sum of AQI
Sum of Temperature Avg (°C)
Level of Detail: City, State
Scaling: Normalized

Summary Diagnostics

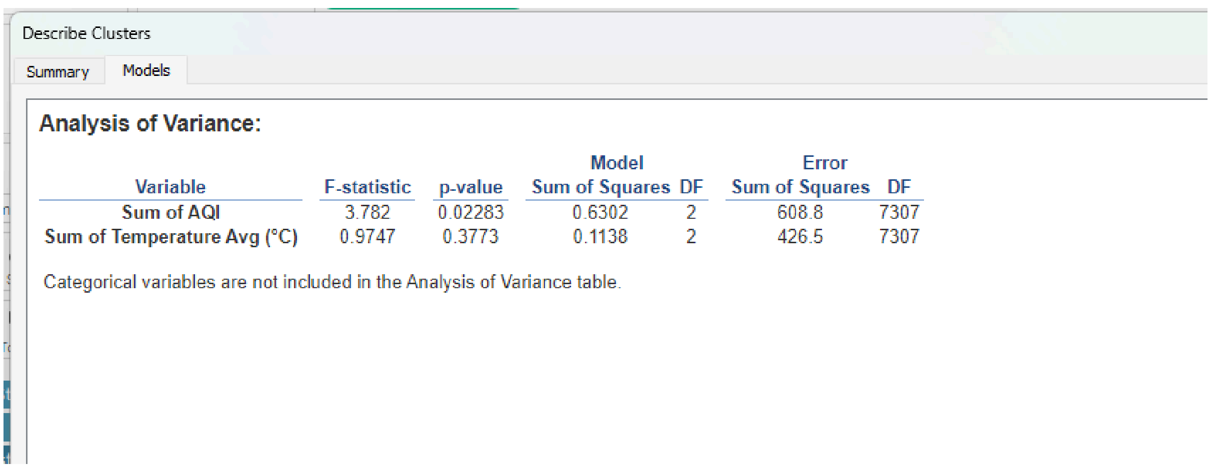
Number of Clusters: 3
Number of Points: 7310
Between-group Sum of Squares: 1065.6
Within-group Sum of Squares: 4761.7
Total Sum of Squares: 5827.3

		Centers		Most Common	
Clusters	Number of Items	Sum of AQI	Sum of Temperature Avg (°C)	City	State
Cluster 1	2924	190.57	29.91	Jaipur	Maharashtra
Cluster 2	2193	194.34	30.129	Delhi	West Bengal
Cluster 3	2193	197.44	29.925	Chennai	Tamil Nadu
Not Clustered	0				

Show scaled centers

Copy to Clipboard | Learn more about the cluster summary statistics | Close

Step 4: Navigate to the **Models** view to inspect the ANOVA results and statistical significance markers.



Describe Clusters

Summary | Models

Analysis of Variance:

			Model		Error	
Variable	F-statistic	p-value	Sum of Squares	DF	Sum of Squares	DF
Sum of AQI	3.782	0.02283	0.6302	2	608.8	7307
Sum of Temperature Avg (°C)	0.9747	0.3773	0.1138	2	426.5	7307

Categorical variables are not included in the Analysis of Variance table.

D) Analyze distributions using built-in tools

Task 1: Generate Temperature Distribution Profiles

Step 1: Initialize a fresh worksheet. Assign label: **Temperature Distributions**

Step 2: Pull: **`City`** → **`Columns`**

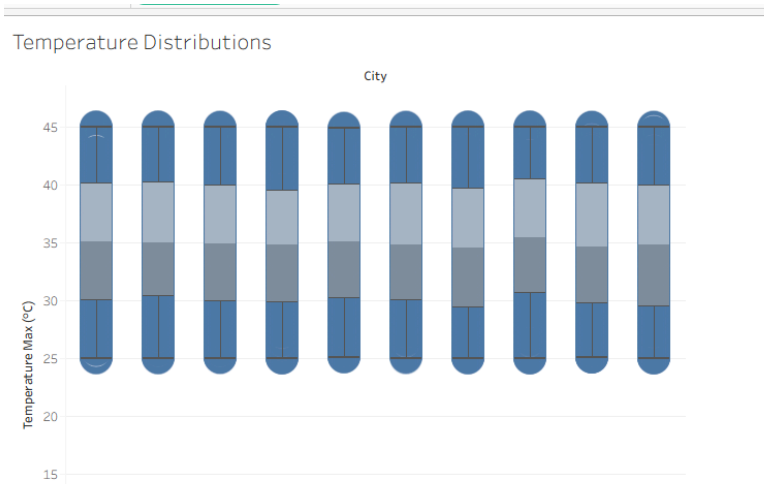
Step 3: Pull: **`Temperature_Max (°C)`** → **`Rows`**

Step 4: Modify **`Temperature_Max`** to **Dimension** or navigate to: **Analysis → deselect Aggregate Measures**

Step 5: Access the **Analytics** window.

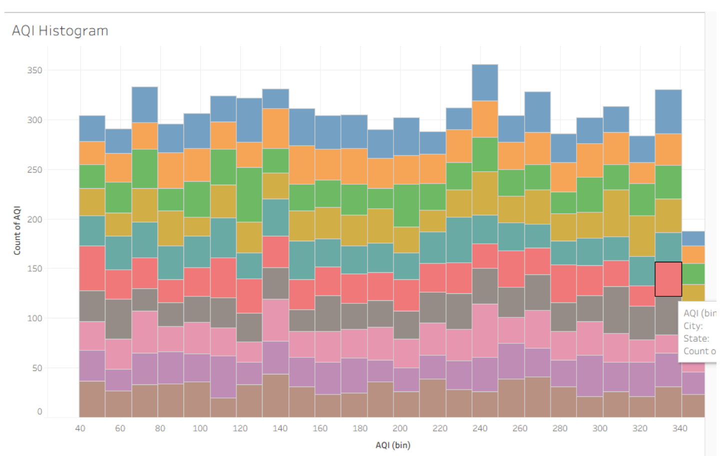
Step 6: From the **Custom** section, move **`Box Plot`** directly onto the workspace.

Result: A visual representation showcasing temperature spreads across cities, highlighting median values, spread, and outlier data points.



Task 2: Develop an AQI Histogram

Step 1: Initialize a fresh worksheet. Assign label: **AQI Histogram**
Step 2: Pick the following measure: **`AQI`** within the Data pane.
Step 3: Access the: **Show Me**
Step 4: Navigate to and select: **`Histogram`** The software will generate frequency bins for AQI automatically.
Step 5: Pull the following dimension: **`City → Color`**
Result: A chart depicting the concentration and spread of pollution levels across various urban centers.

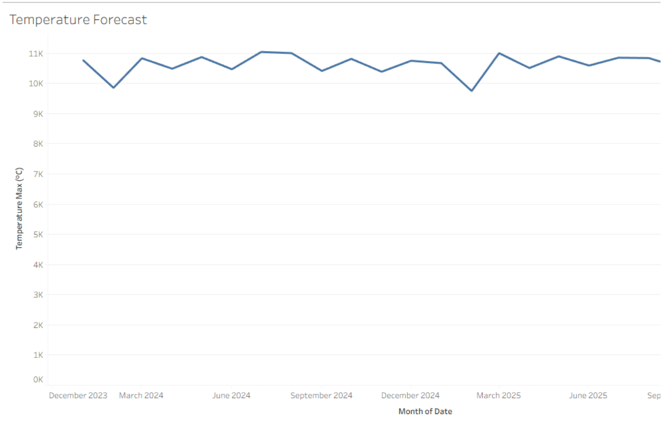


E) Apply forecasting techniques

Task 1: Generate Temperature Forecast Models

Step 1: Initialize a fresh worksheet. Assign label: **Temperature Forecast**
Step 2: Pull: **`Date → Columns`**
Step 3: Access the dropdown menu on the Date pill. Navigate to and select: **Continuous Month**
Verify that the axis reflects chronological intervals such as: **`Jan 2024, Feb 2024...`**

Step 4: Pull: **`Temperature_Max (°C)` → `Rows`**
Step 5: Ensure the measure is configured as: **`AVG(Temperature_Max (°C))`**

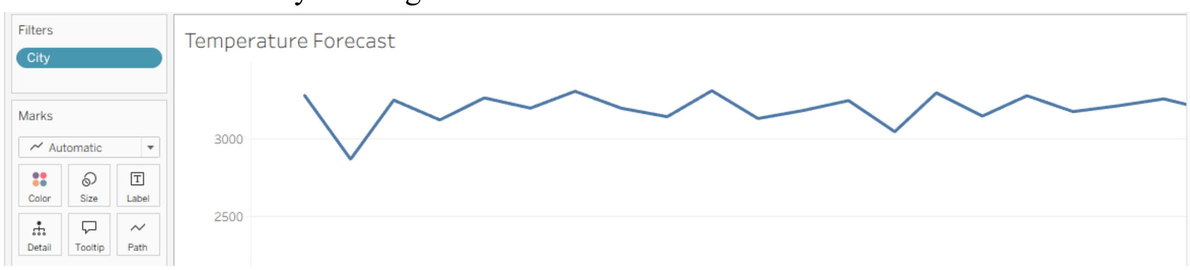


Task 2: Refine the Geographical Scope

Step 1: Move the following dimension: **`City` → `Filters`**
Step 2: Choose a subset of urban areas (e.g., 2–3 locations):

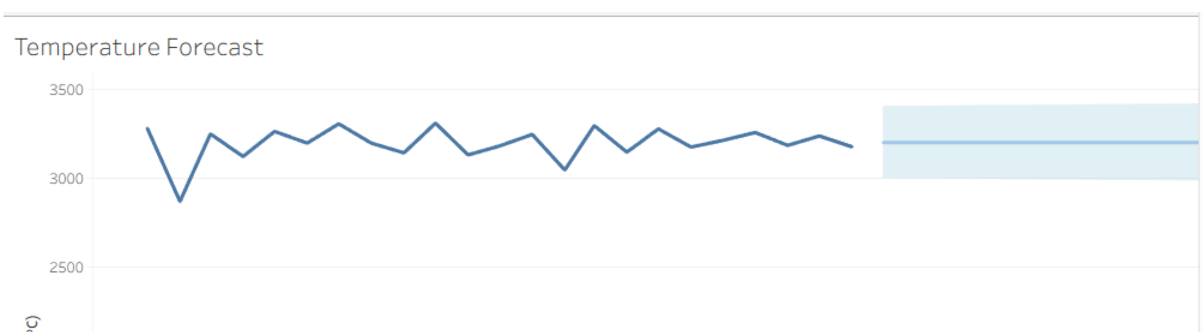
- Mumbai
- Delhi
- Bengaluru

Confirm the selection by clicking **OK**.



Task 3: Implement Predictive Forecast Models

Step 1: Enter the Analytics window.
Step 2: From the Model section, pull: **`Forecast`** directly onto the workspace.
The application will project the visualization into future periods using a calculated prediction interval.



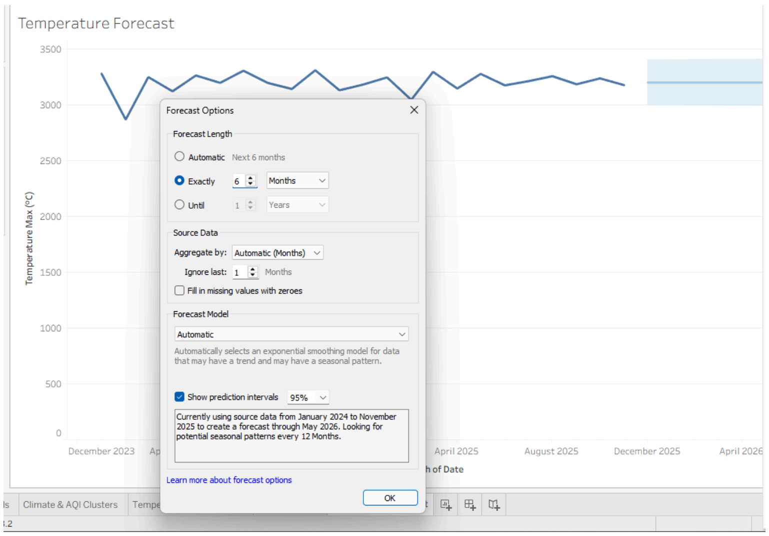
Task 4: Tailoring the Predictive Forecast

Step 1: Right-click within the visualization area. Navigate to the following menu: **`Forecast → Forecast Options`**

Step 2: Adjust the configuration to: **Forecast Length = Exactly 6 Months**

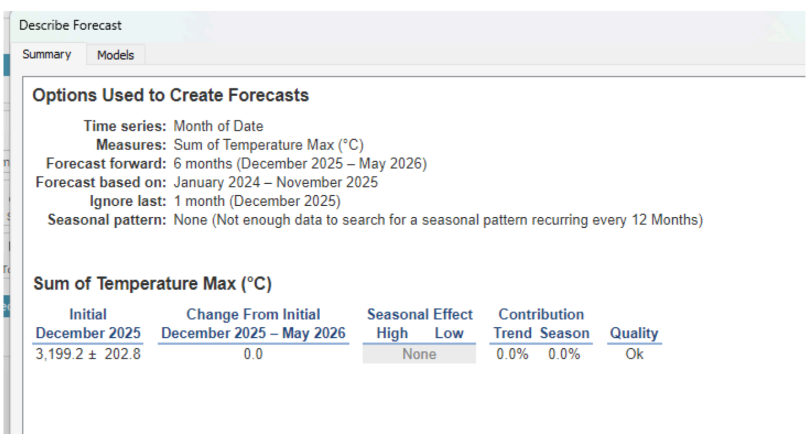
Step 3: Configure the following parameter: **Prediction Interval = 95%** Confirm the changes by clicking **OK**.

Result: The plot now reflects projected maximum temperatures for the subsequent half-year period.



Task 5: Evaluating the Predictive Model

Step 1: Right-click within the visualization area. Navigate to: **`Forecast → Describe Forecast`** This action presents a detailed breakdown of the statistical results.



Describe Forecast

Summary | Models

Options Used to Create Forecasts

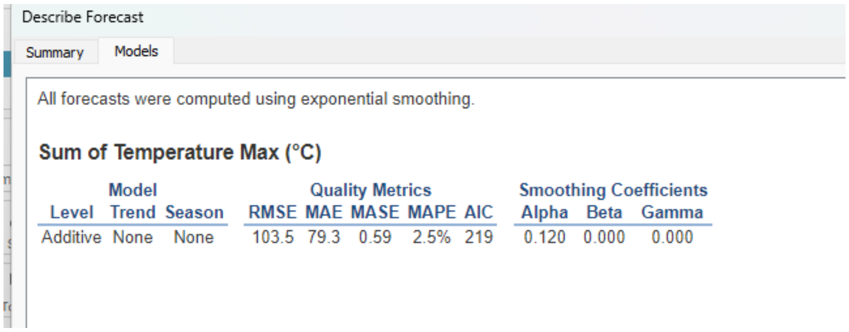
Time series: Month of Date
Measures: Sum of Temperature Max (°C)
Forecast forward: 6 months (December 2025 – May 2026)
Forecast based on: January 2024 – November 2025
Ignore last: 1 month (December 2025)
Seasonal pattern: None (Not enough data to search for a seasonal pattern recurring every 12 Months)

Sum of Temperature Max (°C)

Initial	Change From Initial	Seasonal Effect		Contribution		
December 2025	December 2025 – May 2026	High	Low	Trend	Season	Quality
3,199.2 ± 202.8	0.0	None		0.0%	0.0%	Ok

Step 2: Access the **Models** view. Inspect the following performance indicators:

- Root Mean Square Error (RMSE)
- Selected Model Configuration
- Trend Characteristics
- Seasonal Pattern Analysis


Describe Forecast

Summary | Models

All forecasts were computed using exponential smoothing.

Sum of Temperature Max (°C)

Model			Quality Metrics					Smoothing Coefficients		
Level	Trend	Season	RMSE	MAE	MASE	MAPE	AIC	Alpha	Beta	Gamma
Additive	None	None	103.5	79.3	0.59	2.5%	219	0.120	0.000	0.000

F) Compare different statistical models

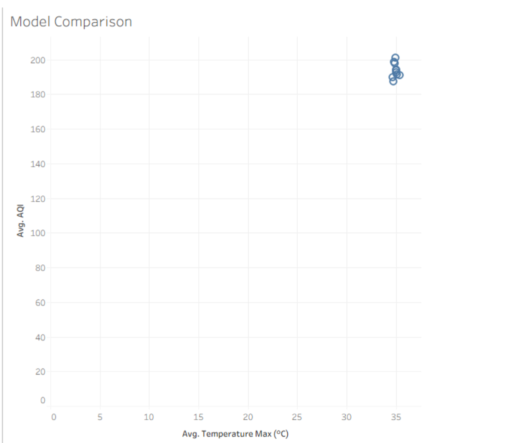
Task 1: Generate Model Comparison Profiles

Step 1: Initialize a fresh worksheet. Assign label: **Model Comparison**

Step 2: Pull the following measure: **`Temperature_Max (°C)`** → **`Columns`** Configure the aggregation to **`AVG`**.

Step 3: Pull the following measure: **`AQI`** → **`Rows`** Ensure the field is set to **`AVG`**.

Step 4: Move the following dimension: **`City`** → **`Detail`** onto the Marks card.



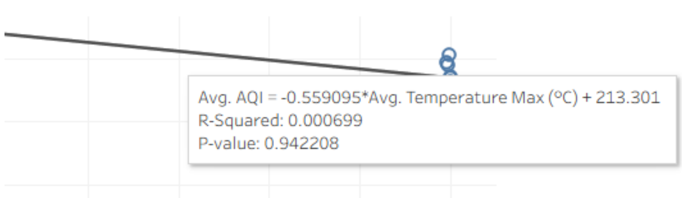
Task 2: Implement and Evaluate Predictive Models

Step 1: Enter the Analytics window.

Step 2: From the Model section, pull: **`Trend Line`** → **`Linear`**

Step 3: Position the pointer over the resulting line and record:

- R-squared coefficient
- Probability value (P-value)

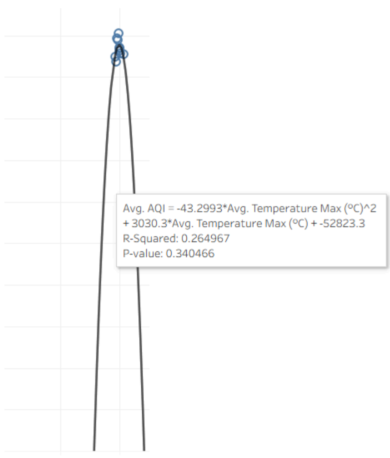


Step 4: Perform a right-click on the trend line. Navigate to the following menu: **`Edit Trend Lines`**

Step 5: Assess the following model types:

- Linear
- Logarithmic
- Exponential
- Polynomial of Second Degree
- Polynomial of Third Degree

Step 6: Evaluate the corresponding **R-squared indicators**. The configuration exhibiting superior R-squared metrics and a **P-value under 0.05** typically indicates valid statistical correlation.



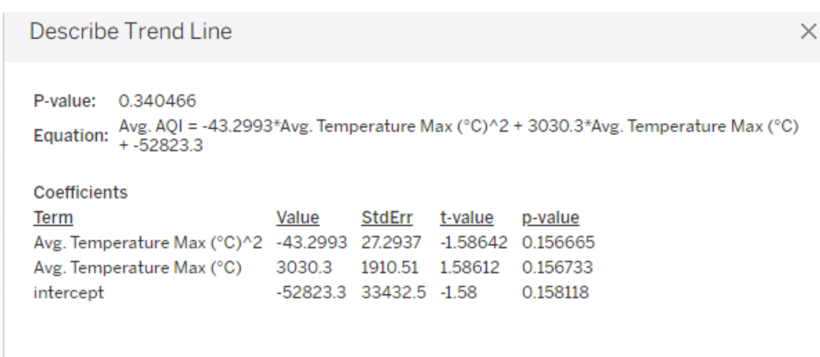
G) Export and interpret statistical results

Task 1: Exporting the Statistical Trend Model

Step 1: Navigate to the worksheet labeled: **Model Comparison**

Step 2: Perform a right-click on the trend line. Choose the following option: **`Describe Trend Model`**

Step 3: Select either: **Copy** or **Export** to preserve the mathematical output as an external document.



Describe Trend Line

P-value: 0.340466

Equation: Avg. AQI = -43.2993*Avg. Temperature Max (°C)^2 + 3030.3*Avg. Temperature Max (°C) + -52823.3

Coefficients

Term	Value	StdErr	t-value	p-value
Avg. Temperature Max (°C)^2	-43.2993	27.2937	-1.58642	0.156665
Avg. Temperature Max (°C)	3030.3	1910.51	1.58612	0.156733
intercept	-52823.3	33432.5	-1.58	0.158118

Task 2: Evaluate and Document Statistical Metrics

Document the following three indicators from the generated Tableau workspace:

1. Mathematical Formula Components

Observe the following:

- Y-intercept value

- Slope/regression coefficient

2. R-squared Coefficient

This metric defines the proportion of variance in the target field clarified by the model. For instance: $R^2 = 0.74$ indicates that **74% of observed variations are accounted for by the model**.

3. Probability Value (P-value) Should the following occur: $p < 0.05$ The observed correlation is typically deemed statistically valid. The reference documentation designates the equation constants, R^2 metrics, and p-values as the primary statistical outputs for analysis.

Describe Trend Model

Model formula:	(Avg. Temperature Max (°C)^2 + Avg. Temperature Max (°C) + intercept)
Number of modeled observations:	10
Number of filtered observations:	0
Model degrees of freedom:	3
Residual degrees of freedom (DF):	7
SSE (sum squared error):	124.21
MSE (mean squared error):	17.7443
R-Squared:	0.264967
Standard error:	4.2124
p-value (significance):	0.340466

Individual trend lines:

Panes		Line		Coefficients				
Row	Column	p-value	DF	Term	Value	StdErr	t-value	p-value
AQI	Temperature Max (°C)	0.340466	7	Avg. Temperature Max (°C)^2	-43.2993	27.2937	-1.58642	0.156665
				Avg. Temperature Max (°C)	3030.3	1910.51	1.58612	0.156733
				intercept	-52823.3	33432.5	-1.58	0.158118

Conclusion: The practical successfully demonstrated advanced analytics in Tableau using the Indian Climate Dataset. Trend lines, clustering, box plots, histograms, forecasting, and statistical model comparison were used to identify relationships, patterns, distributions, and future climate trends